Apoorv Vuppulury

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Summary

Final-year ECE student with a strong focus on computer architecture and hardware–software co-design, backed by a solid foundation in embedded systems, VLSI design, and system-level modeling. Experienced in embedded control and timing/performance analysis across hardware–software boundaries, with a proven record of executing complex, long-term technical & interdisciplinary projects.

Education

B.E. in Electronics and Communication Engineering Oct, 2022 – Jul, 2026
BITS Pilani - Hyderabad Campus CGPA: 8.01/10.00

- **Coursework:** Computer Architecture, Digital VLSI Design, Machine Learning, Embedded System Design, Digital Design
- **NPTEL:** Computer Architecture and Organization, Hardware Modeling using Verilog, 5G - Wireless Standard Design

Experience

Firmware Engineer - II Jul, 2026 - Present
Dhruva Space Hyderabad, TG

- Part of **Satellite IoT & Data Relay Systems** to support the design and deployment of SmallSat-based IoT data relay solutions. Working on end-to-end IoT systems across space and ground segments and the ability to translate customer use cases into practical satellite-IoT architectures.

Embedded AI Intern Jan, 2026 - Jun, 2026
Inferigence Quotient Bengaluru, KA

- Designed and implemented a cascaded closed-loop 3-axis gimbal control system using Teensy 4.x, featuring interrupt-driven encoder decoding and real-time PID gain tuning for deterministic, low-latency position control.
- Developed a MATLAB/Simulink-based dynamic model of the control loop and validated hardware performance through flight telemetry analysis (QGC), achieving <5% deviation between simulation and experimental results.

Projects

Asynchronous FIFO Memory System | GitHub Repository Apr, 2025 - Oct, 2025

- Designed a clock-domain-crossing (CDC)-based asynchronous FIFO with Gray-coded pointers and dual flip-flop synchronizers to safely transfer data between asynchronous clock domains and mitigate metastability.
- Developed an FSM-based request–acknowledge handshake to enable reliable single-word transfers across the asynchronous buffer.

5 - Stage pipeline RISC Processor | GitHub Repository Jan, 2025 - Apr, 2025

- Designed a 5-stage pipelined RISC CPU, supporting custom (R,I,J-type) instruction opcodes with complete datapath and control logic. Verified pipeline correctness through waveform-based simulation in Xilinx Vivado.
- Implemented hazard detection and data forwarding units to resolve pipeline data hazards and minimize stall cycles.

ML and QoS-Aware Routing for LEO Satellite Networks | Apr, 2025 - Nov, 2025

- Designed a QoS-aware routing framework for dynamic LEO inter-satellite networks using Graph Neural Networks (GNNs) for topology embedding and Deep Q-Networks (DQN) for adaptive path selection under time-varying link conditions.
- Achieved up to 35% latency reduction and improved QoS satisfaction by 8% compared to Dijkstra and CGR baselines.

Embedded Edge-Framework for Autonomous Collision Avoidance | Aug, 2025 - Dec, 2025

- Developed an embedded vision platform on Raspberry Pi and NVIDIA Jetson Nano with cameras for onboard autonomous obstacle detection and collision avoidance. Implemented an OpenCV-based image processing pipeline to detect obstacles from live video and trigger evasive action.

Resource Allocation in STAR-RIS Assisted Wireless 6G Networks | Oct, 2024 - Dec, 2025

- Investigated resource allocation strategies, deriving mathematical bounds and critical trade-offs on element allocation factors and phase error constraints to optimize spectral efficiency. Developed a three-stage framework for intelligent coverage hole detection and remediation in PPP networks that achieved 100% coverage, while improving baseline coverage by 12.28%.

Technical Skills

- Tools: Xilinx Vivado, MATLAB (Simulink, Simscape), Arduino IDE, STM32Cube IDE, Keil MDK-ARM
- Languages: Verilog HDL, ARM Assembly, MATLAB, C, C++, Python (NumPy, SciPy, TensorFlow, Pandas, MatPlotLib)
- Communication Protocols: UART, SPI, I2C, MavLink, Ethernet, WIFI (IEEE 802.11), HTTP, TCP, UDP
- Embedded Platforms: Arduino, ESPs, Teensy 4.x, Raspberry Pi, STM32, NVIDIA Jetson Nano, FOC Shields

Positions of Responsibility

Founding Member, Training and Counseling Lead | **S**tudent **W**elfare and **M**entorship **C**ommittee May, 2024 – Aug, 2025

- Designed the first comprehensive mentorship program of BPHC, and led a team of 150 student mentors, positively impacting over 500 first-year students and addressing key transitional challenges from Day 0.
- Served as a member of the Student–Faculty Council and received a Certificate of Appreciation for contributions to the Dept. of EEE.